Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/07/2006
PI Name:	Kronenberg, Amy Sc.D.		
Project Title:	Comparative Analysis of Fe-Induced Autosomal Mutations in Murine Tissue and Cell Lines		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-8099	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	05/19/2003	**End Date:**	03/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 03/31/2008 per PI (6/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Turker, Mitchell (Oregon Health and Sciences University) Gauny, Stacey (Lawrence Berkeley National Laboratory) Connolly, Lanelle (Oregon Health and Sciences University) Dan, Cristian (Oregon Health and Science University) Kwoh, Ely (Lawrence Berkeley National Laboratory)		
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Performance Goal No.:			
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Task Description:	Cancer induction in humans involves a variety of genetic and epigenetic alterations, including widespread loss of expression of genes located primarily on autosomes. Currently, no in vivo data are available to determine the risks of mutagenesis induced by space radiations at an autosomal locus. This study will determine the mutagenic risks of Fe ions using a model autosomal locus (Aprt) in the B6D2 mouse. Four aims are proposed: Aim 1 tests the hypothesis that Fe ions induce autosomal mutations at the Aprt locus in B6D2 kidney epithelial cells exposed in vivo. Aim 2 tests the hypothesis that the exposure to Fe ions elicits primarily multi-locus deletions inclusive of Aprt in B6D2 kidney epithelial cells exposed in vivo. Aim 3 compares Fe-induced mutant frequencies in B6D2 kidney epithelial cells exposed in vivo or in vitro to test the hypothesis that the primary determinant of mutagenic risk is the energy deposited in the cells and not potential modulating effects of the cellular microenvironment. Aim 4 tests the hypothesis that the primary determinant of mutational spectrum is the energy deposited in the cells at risk by comparing the spectrum of Aprt mutants arising after in vitro exposure of B6D2 kidney epithelial cell lines to Fe ions with the spectrum of Aprt mutants obtained after in vivo exposure of kidney epithelial cells (Aim 2). Most literature on the biological effects of heavy ions describes results obtained with in vitro assays, yet considerable uncertainty exists as to how well these assays predict in vivo responses, particularly for endpoints relevant to carcinogenesis. Upon completion of the proposed work, NASA should be in a better position to determine whether cellular assays can be relied upon with a high degree of confidence, or whether greater emphasis should be placed on the establishment and testing of animal models. The goal of this study is to determine whether the tissue microenvironment modifies the mutagenic response of epithelial cells to charged particle radiation of importance in spaceflight. Mutant frequencies and mutant spectra will be evaluated in isogenic kidney epithelial cells exposed either in vitro or in vivo. Little information is available on the mutagenic effects of charged particle radiations in epithelial cell systems. The primary goal of this work is to establish such data and to determine how or if the cellular microenvironment modulates the expression of mutagenic damage.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These studies will be useful for the assessment of the importance of the cellular microenvironment on the frequency and types of mutations induced at autosomal loci following exposure to ionizing radiation, with a particular emphasis on densely ionizing radiations such as alpha particles. These studies may also be of interest to the medical community as it considers the use of charged particles more widely in radiotherapy applications.
Task Progress:	Progress continued on both our in vitro and in vivo experiments designed to assess the impact of the tissue microenvironment on the cytotoxic and mutagenic properties of densely ionizing radiation in an epithelial cell type. We continue to utilize Aprt heterozygous B6D2 F1 mice, or kidney cells derived therefrom. During the past year we completed the assessment of in vitro mutagenesis following exposure of a particular clone of Aprt heterozygous kidney epithelial cells to 1 GeV/amu Fe ions with additional experiments during NSRL5 (Spring 2005). The results confirm a linear dose-response for the induction of autosomal mutations in an endogenous locus in an epithelial cell model. This is consistent with what we had previously observed for the TK1 locus in human lymphoid cells. Also during this past year we developed additional primary kidney epithelial cells for characterization of cytotoxicity and mutation induction at the Aprt locus. It is planned that these cells will be used for replicate toxicity experiments and preliminary mutation experiments in the Spring 2006 NSRL run (March/April 2006). Our in vivo experiments proceeded with completion of the data analysis from the first cohort of irradiated mice (NSRL2, Spring 2004) for cells harvested 8-10 months post-irradiation. These results confirmed our observation of a dose-dependent decrement in cell viability that was seen in the early harvest animals (2-4 months post-irradiation), and indicate that the cytotoxic effects of Fe ion exposures can be detected for a long time after the exposure. This is consistent with previous work by others using morphological assessments of kidney tubule repopulation, that indicate that the kidney is a late responding organ following exposure to sparsely ionizing radiation such as x-rays. Characterization of the mutagenic response for kidney cells exposed in vivo and incubated in situ continued with the exposure of the second cohort of animals in Spring 2005 (NSRL5). It remains premature to describe the in vivo dose-response relationship, but the results from the second cohort, early harvest animals, confirms that Fe ions elicit a mutagenic response in the range of (0-2 Gy) exposure that can be readily differentiated from the control levels in parallel, unirradiated animals. We are presently completing the mutant frequency analysis on the late harvest animals from NSRL5, and are exposing the next cohort of animals in the Spring 2006 NSRL run. Molecular characterization of Aprt-deficient mutants that arose in kidneys exposed in vivo or in kidney cells exposed in vitro continues. The results to date are consistent with the formation of multilocus deletions and mitotic recombination events that involve the induction of DNA double-strand breaks. As such, these mutants resemble the TK1 mutants we previously characterized from human lymphoid cells exposed in vitro to 1 GeV/amu Fe ions during earlier AGS experiments.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Kronenberg A, Gauny SS, Connolly L, Turker M. "Impact of the tissue microenvironment on cell killing and autosomal mutagenesis in murine cells exposed to 1 GeV/amu Fe ions." 16th NASA Space Radiation Health Investigators' Workshop Proceedings, Port Jefferson, NY, May 2005. 16th NASA Space Radiation Health Investigators' Workshop Proceedings, 2005. , May-2005
Abstracts for Journals and Proceedings	Kronenberg A. "Exploring the roles of physics and biology in mediating charged particle radiation responses in mammalian cells and tissues." Proceedings of the European Radiation Research Society Meeting, Leicester, England, September 2005. Proceedings of the European Radiation Research Society Meeting, 2005. , Sep-2005
Abstracts for Journals and Proceedings	Kronenberg A, Gauny SS, Connolly L, Turker M. "Impact of the tissue microenvironment on cell killing and autosomal mutagenesis in murine epithelial cells exposed to 1 GeV/amu Fe ions." Proceedings of the 52nd Annual Meeting of the Radiation Research Society, Denver, CO, October 2005. Proceedings of the 52nd Annual Meeting of the Radiation Research Society, 2005. , Oct-2005
Articles in Peer-reviewed Journals	Durante M, Kronenberg A. "Ground-based research with heavy ions for space radiation protection." Adv Space Res. 2005;35(2):180-4. PMID: 15934192 , Jul-2005

Presentation	Kronenberg, A. "Exploring the roles of physics and biology in mediating charged particle radiation responses in mammalian cells and tissues. " European Radiation Research Society Meeting, Leicester, England Sep-2005
Presentation	Kronenberg, A. "Charged Particle Radiation and the Apoptotic Response" NASA Summer School in Radiobiology Jun-2005
Presentation	Kronenberg, A. "Charged Particle Radiation-Induced Mutagenesis and Genomic Instability" NASA Summer School in Radiobiology Jun-2005
Presentation	Kronenberg, A. "Mutagenesis and Genomic Instability" Loma Linda University University Course for Radiation Oncology Residents May-2005
Presentation	Kronenberg, A. "Apoptosis" Loma Linda University Medical School Radiation Oncology Residents Course May-2005